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Printed Pages : 3

EBM801

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 101804

Roll No.

B. Tech.

(SEM. VIII) THEORY EXAMINATION, 2014-15

COMMUNICATION ENGINEERING

Time : **3** Hours] [Total Marks : **100**

1 Attempt any four parts of the following : **5×4=20**

(a) What is meant by the term amplitude modulation ?

(b) Define the form modulation index for AM.

(c) Derive the power relations for single-tone amplitude modulated wave.

(d) Derive an expression for the transmission efficiency of AM wave.

(e) Draw the block diagram of a superheterodyne receiver and explain the function of each block.

(f) How can you classify frequency spectrum of FM wave ?

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2 Attempt any four parts of the following : **5×4=20**

(a) What are the advantage and applications of pulse-code modulation ?

(b) Explain the sampling theorem.

(c) Explain generation and demodulation of pulse-position modulation (PPM).

(d) What is the fundamental difference between pulse modulation, on the one hand, and frequency and amplitude modulation on the other ?

(e) Explain why quantizing noise could affect small amplitude signals in a PCM system.

(f) A system has a bandwidth of 4 kHz and a signal-to-noise ratio of 28 dB at the input to the receiver. Calculate :

(i) its information carrying capacity

(ii) the capacity of the channel if its bandwidth is doubled. while the transmitted signal power remains constant.

3 Attempt any two parts of the following : **10×2=20**

(a) Draw and explain the working of microwave radio system with its advantage.

(b) Explain the following :

(i) Look angle

(ii) Satellite link model

(c) Explain what is meant by saying that a satellite is "stationary" why are such satellite used for worldwide communications in preference to any kind ?

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4 Attempt any two parts of the following : **10×2=20**

(a) Define and explain information and information theory. What are the aims of information theory ? Why is meaning divorced from information ?

(b) Explain the following :

(i) Information Rate

(ii) Channel Capacity

(c) Describe three kind of error-detection code and explain how they detect data errors.

5 Attempt any two parts of the following : **10×2=20**

(a) Explain the elements of cellular mobile radio-Handoff frequency management and channel assignment.

(b) Explain the significance of basic cellular concept and its operation.

(c) Explain the following :

(i) GSM

(ii) WCDMA